

Adversarial Cooperation

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Abstract—Adversarial Cooperation seeks solutions to cooperation problems using cryptographic protocols without a trusted referee, and hardware for their execution. The aim is cooperation that preserves private information and the strategic advantages it supports. We seek resources to develop, analyze, implement, and demonstrate these protocols and their hardware.

Index Terms—Applied cryptography, cooperation, zero knowledge, multiparty computation, hardware.

I. THE COOPERATION PROBLEM

The central question is: *how can adversarial parties cooperate without a referee while retaining their strategic advantage?* Parties should verify an agreed result without surrendering cards, strategies, or other secrets. We are concerned that public and private institutions may underestimate the information their interactions expose.

The privacy objective is to reveal nothing about private inputs beyond the agreed outputs and public information. Each protocol must specify its adversary model, assumptions, and privacy and correctness requirements.

The program builds on commitments, zero-knowledge proofs, and multiparty computation [1]–[3], alongside prior work on replacing mediators in strategic games [4]. No new primitive or general solution is claimed; each proposed construction requires its own analysis.

II. GAMES AND COOPERATION PROBLEMS

The following cases define initial research targets:

- *Rock–Paper–Scissors*: Fix each player’s choice before either learns the other’s move, then verify the completed round’s outcome.
- *Tic-Tac-Toe*: Demonstrate that a private strategy avoids defeat under specified game conditions, without disclosing its move-selection rules.
- *Poker*: Investigate dealing and winner verification without a trusted dealer, without revealing private cards beyond what the agreed outcome implies.
- *Chess*: Determine the outcome between fixed, private deterministic move-selection algorithms under agreed rules and computation bounds, without publishing their moves or strategies.
- *Stag Hunt*: Coordinate mutually beneficial action when each party’s willingness to cooperate depends on the other’s participation.
- *Joint Optimization*: Optimize an agreed objective using private inputs or constraints, revealing only the agreed result.

Each case requires its own specification and analysis. Whether more complex cooperation can be composed from simpler, elemental games and protocols remains an open question.

III. FROM SPECIFICATION TO HARDWARE

The goal is to take formally defined protocols through inspectable executable representations into specialized hardware that participants can independently inspect and reproduce. Candidate methods include C reference code, Boolean circuits, register-transfer-level (RTL) descriptions in Verilog or VHDL, and FPGA evaluation; choices depend on the protocol.

Translation correctness and correspondence between specification, code, circuits, and devices are research tasks. Prior work on verified high-level synthesis provides a foundation [5]. Evaluation must address functional behavior, costs, fabrication and device integrity, and physical leakage or faults. Functional correspondence alone does not establish protection against such physical attacks.

IV. RESOURCES AND SUPPORT

We seek funding, research collaborators, computing resources, FPGA boards, and measurement equipment. Support would enable dedicated research, independent cryptographic review, implementation, and hardware experiments.

For each selected cooperation problem, the aim is a complete, inspectable demonstrator: protocol definition, implementation, adversarial evaluation, and hardware realization. Proof status and limitations will be explicit; a working demonstrator does not establish a cryptographic proof or general composability.

For funding, equipment access, or collaboration, contact contact@waajacu.com. Peace is the ethical motivation, not a property a cryptographic proof can establish.

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